

Xinyu Jia

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EDUCATION

National University of Singapore

Ph.D. in Engineering (Robotics)

Thesis: Efficient Motion Control via Estimation and Optimization for Robotic Manipulation

Singapore, Singapore

2020.08 - 2025.01

Jilin University

B.Eng. in Vehicle Engineering

Changchun, China

2014.09 - 2018.06

RESEARCH INTERESTS

Optimization, Model Predictive Control, Nonlinear Control, Robust Control, Motion Planning, Motion Control, Robot Learning, Robot Perception, Robot Safety, Robot System Design, Mechatronics

SKILLS AND EXPERTISE

Programming: C++, Python, MATLAB

Libraries and Tools: PyTorch, NumPy, SciPy, Stable-Baselines3, Gymnasium, Transformers, LeRobot, Miniconda | Pinocchio, Crocoddyl, CasADi, QuadProg, qpOASES, IPOPT, Eigen, ViSP | ROS, Gazebo, PyBullet, CoppeliaSim, MoveIt | CMake, pthreads, CAN, UDP, TTL, Arduino | CATIA, AutoCAD, Adams/Car, ANSYS, Altium Designer

Languages: Chinese (native), English (fluent)

WORK EXPERIENCE

Shenzhen ZYT Technology Co., Ltd., China (DJI Automotive)

Algorithm Engineer (Intern)

2025.02 - 2025.05

- Investigated, developed and deployed vision-language-action models for robot systems.
- Developed reinforcement learning algorithms for motion control of robotic manipulators.

NUS Advanced Robotics Centre, Singapore

Research Engineer

2022.04 - 2023.04

- Developed novel motion planning and control algorithms for high-degree-of-freedom robots.
- Designed and built an 18-DoF quadruped manipulator for dynamic loco-manipulation ([link](#)).
- Designed and developed a 12-DoF bimanual cobot for physical human-robot interaction ([link](#)).

NUS Advanced Robotics Centre, Singapore

Research Assistant

2021.07 - 2022.03

- Developed a 12-DoF, 16.5 kg, electrically actuated, torque-controllable quadruped robot.
- Designed and developed a 6-DoF, 3.5 kg robotic arm with 1 kg payload at 0.6 m maximum reach.

Shenzhen DJI Technology Co., Ltd., China

Mechanical Engineer

2018.07 - 2020.06

- Developed hardware of an electronic control unit and a stereo camera for autonomous driving.

Shenzhen DJI Technology Co., Ltd., China

Chassis Engineer (Intern)

2018.01 - 2018.05

- Analyzed and evaluated vehicle chassis performance via numerical simulation.

COMPETITION EXPERIENCE

2018 Formula Student China (FSC 2018)

Technical Consultant

2017.12 - 2018.06

- Provided technical direction for Gspeed Formula Racing Team ([link](#)).
- Led the design and development of a carbon-fiber-reinforced-polymer (CFRP) monocoque chassis.

2017 Formula Student China (FSC 2017)

Technical Leader

2016.12 - 2017.11

- Led the development of a racing chassis including suspension, steering, and braking systems ([link](#)).
- Proposed and built a novel third-spring suspension system, enhancing vehicle handling stability.

2016 Formula Student China (FSC 2016)

Head of a Suspension Group

2015.12 - 2016.11

- Led the development of a suspension system involving simulation, design, integration, and testing.

2015 Formula Student China (FSC 2015)

Member of a Suspension Group

2014.12 - 2015.11

- Designed, optimized, and implemented a lightweight suspension upright structure for a race car.

PUBLICATIONS

- **X. Jia**, W. Wang, J. Yang, Y. Pan and H. Yu, “ToMPC: Task-oriented Model Predictive Control via ADMM for Safe Robotic Manipulation,” *IEEE Robotics and Automation Letters*, 2025.
- **X. Jia**, W. Wang, J. Yang, Y. Pan and H. Yu, “Multi-layered Safety of Redundant Robot Manipulators via Task-oriented Planning and Control,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- **X. Jia**, J. Yang, T. Shi, W. Wang, Y. Pan and H. Yu, “Robust Precision Motion Control based on Enhanced Unknown System Dynamics Estimator for High-DoF Robot Manipulators,” *IEEE/ASME Transactions on Mechatronics*, 2024.
- **X. Jia**, J. Yang, K. Lu, Y. Pan and H. Yu, “Enhanced Robust Motion Control based on Unknown System Dynamics Estimator for Robot Manipulators,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- J. Yang, **X. Jia**, Z. Hou, Y. Pan and H. Yu, “Robust Reference Tracking of Linear Uncertain Systems via Uncertainty Estimation and Composite Control,” in *IEEE Conference on Decision and Control (CDC)*, 2023.
- K. Lu, S. Han, **X. Jia** and H. Yu, “Inverse Optimal Adaptive Prescribed Performance Control With Application to Compliant Actuator-Driven Robot Manipulators,” in *IEEE Conference on Decision and Control (CDC)*, 2023.

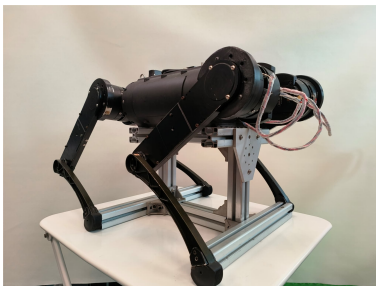
HONORS AND AWARDS

- **Competitions:** National Champion of FSCC 2018, First Prize of FSC 2017, Second Prize of FSC 2016
- **Academics:** Third Class Scholarship & Individual Scholarship in 2018, Third Class Scholarship in 2015

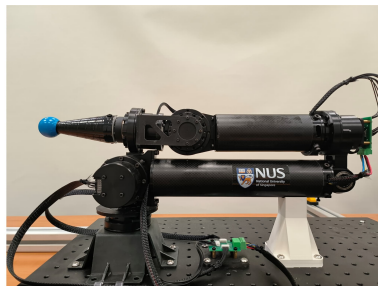
INVITED TALKS

- “Multi-layered Safety of Redundant Robot Manipulators” (NUS Graduate Seminar, 2024.10.30).
- “Motion Control Methods for High Degree of Freedom Robots” (TechBeat, [video](#), 2024.06.13).
- “Locomotion Control for a Quadruped Robot” (NUS Graduate Seminar, 2021.08.25).

ROBOTS I BUILT



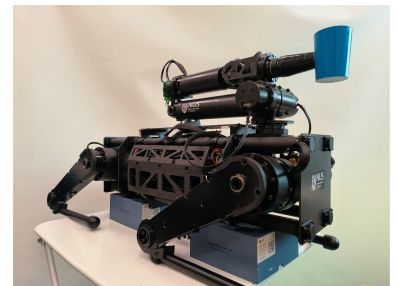
Quadruped Robot



Robotic Manipulator



Bimanual Cobot



Quadruped Manipulator